

NURMEREY AIDOS



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EXPERIENCE

LARCASE, École de technologie supérieure ÉTS

Intern — Aircraft Anomaly Detection

Montréal, Canada

May 2025 - September 2025

- Analyzed **12 Machine Learning algorithms** for flight-test data and addressed the **imbalanced classification problem** with a **9:1 class ratio** using F1 score as the evaluation metric.
- Developed an **ML model for aircraft trim-condition detection** by analyzing **1,286 flight tests** and detecting **candidate stable intervals beyond existing labels**.
- Improved **trim-interval detection F1 score to 0.65** using **CatBoost** and a **sliding-window approach**, achieving a 34% improvement over the baseline model.
- Reduced **overfitting** and optimized model performance using **k-fold cross-validation, early stopping, and Grid Search/Random Search** hyperparameter tuning.

Nazarbayev University

Research Assistant — Polyurethane Nanocomposites and Coatings

Astana, Kazakhstan

March 2025 - May 2026

- Achieved uniform dispersion** of hydrophobic nanofibers in a polyurethane matrix via **ball milling (2 hours, with toluene)**, resolving persistent **nanofiber agglomeration** issues
- Manufactured and mechanically **tested 268 samples** (0–0.5 wt% PNF/PU) under **tensile and shear loading**
- Delivered a **217% increase in tensile strength** and **185% increase in shear strength** through optimized nanofiber loading
- Fabricated **fiber metal laminates** and **polyurethane protective coatings**; conducted mechanical and environmental testing, generating structured datasets for **machine learning model development**

TOO KazAeroSpace

Intern

Astana, Kazakhstan

Summer 2024

- Designed and 3D-printed** functional parts for the "Sky Photography Using a Portable Camera for Aerospace Objects" project, enabling successful **manufacturing and deployment** of the device
- Led **presentations and partnership negotiations with Google** to mount cameras on vehicles for "Google Street View coverage of Kazakhstan", resulting in a **successfully launched program** with cameras mounted

EDUCATION

École de technologie supérieure ÉTS, Montréal, Canada

Master of Science in Aerospace Engineering

Enrolled: 2026 — Expected: 2028

Nazarbayev University, Astana, Kazakhstan

Bachelor of Engineering in Mechanical and Aerospace Engineering

2022 — 2026

Overall GPA: 3.27

Relevant Courses: Aerodynamics, Computer Aided Engineering, Thermodynamics, Machine Elements Design, Control Systems, Vehicle Propulsion Systems, Aerospace Propulsion, Heat Transfer, Mechanical Systems Design.

SKILLS

- Programming:** C++, Matlab, Python, Machine Learning, Pandas, NumPy, SQL
- Software:** ANSYS, SolidWorks, Simulink, Microsoft Office, Wolfram Mathematica
- Certificates:** Certified SolidWorks Associate (CSWA), CITI Program "Social and Behavioral Research"
- Languages:** Kazakh (Native), Russian (Native), English (Advanced), French (Intermediate)

ACHIEVEMENTS

Kazakhstan National Physics Olympiad: **3rd Place**, The Presidential Olympiad in Mathematics, Physics, Chemistry and Biology: **Bronze Medal**, MathCombat by University Association in Mathematics: **1st Place**

PROJECTS AND PUBLICATIONS

Aidos, N., Nurgaziyeva, M. & Zhenis, B. (2026). Multifunctional Coating for Metal/Polymer Surfaces using Polyurethane/Nanofibre Composites. Nazarbayev University School of Engineering and Digital Sciences

In Progress:

Nurmerey Aidos, Mouhamadou Wade, Georges Ghazi, Ruxandra M. Botez, "Aircraft Trim Condition Detection using Flight Test Data and Machine Learning", *AIAA Aviation Forum 2026*.